

AMITH C KOWSHIK

Bangalore, Karnataka

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Education

Bangalore Institute of Technology

Bachelor of Engineering in Computer Science and Engineering
CGPA: 8.25/10

Dec. 2020 – Jun. 2024

Bangalore, India

Experience

F5 Networks

Jul. 2024 – Present

Software Development Engineer II

Bangalore

- Engineered network defense and private cloud products using Golang microservices, enabling distributed deployments across large-scale clusters while incorporating system design principles.
- Delivered a managed DDoS protection platform for enterprise customers, strengthening resilience against large-scale attacks and improving service scalability.
- Processed millions of requests per day with greater than 99.99% uptime, ensuring reliability under heavy traffic.
- Implemented internal monitoring and observability tools, accelerating incident detection by approximately 30%.
- Automated monitoring pipelines, reducing false positives by approximately 20% and enhancing visibility across engineering teams.

F5 Networks

Feb. 2024 – May 2024

Technical Intern

Bangalore

- Created a proof of concept using advanced machine learning techniques to forecast company costs, improving accuracy over traditional time-series models.
- Reduced RMSE and MAE by approximately 20–25% while achieving higher R^2 scores compared to baseline regression models.
- Achieved a 20–30% improvement in prediction efficiency, supporting more reliable financial planning and budgeting decisions.

Fidelity Investments

May 2023 – Jul. 2023

Software Development Engineer Intern

Bangalore

- Built an internal dashboard using Angular, TypeScript, and Tailwind CSS to centralize status monitoring and key details of 30+ company applications.
- Integrated quick-access links and essential data views, streamlining access to critical information.
- Supported access to application information for more than five engineering teams.

Publications

Crowd Control and Monitoring using Deep Learning | *Computer Vision*

Oct. 2024

[IEEE Xplore Link](#)

- M S Janardhana Reddy, Amith C Kowshik, Chakshu Ramaswamy Sharma, Aayush, K J Bhanushree. Published in IEEE, Oct. 2024.
- Designed a deep-learning-based crowd monitoring system using YOLOv8-Pose for human pose detection, social distancing analysis, and crowd-density estimation.
- Integrated automated homography techniques to transform 2D pose detections into 3D coordinates, improving distance estimation and depth perception.
- Generated color-coded density maps and violation alerts to identify overcrowding and social-distancing violations in real-world environments.
- Achieved up to 98.7% accuracy on evaluated crowd-monitoring scenarios.
- This work won the **1st Prize** at the National Level Project Competition 2024 organized by the Department of Computer Science, Bangalore Institute of Technology.

Projects

On-The-Go Rental System | *Full-Stack*

Jan. 2023 – Feb. 2023

- Built a full-stack bike rental platform supporting user management, bike and terminal administration, rental workflows, and payment processing.
- Architected the system using four modular Django applications to separate business logic, improve maintainability, and support scalable feature development.
- Containerized the application with Docker Compose, enabling reproducible development and deployment environments.
- Implemented end-to-end payment and transaction management workflows, including rental payments, transaction history tracking, and administrative reconciliation.
- Created automated tests for core models and views to improve application reliability and streamline development.
- **Tech Stack:** Django, Python, MySQL, Docker, Docker Compose.

Technical Skills

Languages: Golang, C, C++, Java, Python, TypeScript, JavaScript

Databases: MySQL, PostgreSQL, Redis

Frameworks & Libraries: React, Angular, Django, Node.js, Tailwind CSS

Cloud & Infrastructure: Docker, Kubernetes, AWS

Monitoring & Observability: Prometheus, Grafana, Alertmanager

Version Control: Git

Extracurricular

Diminished 7th – Music Club, Bangalore Institute of Technology

Jun. 2022 – Jun. 2024

President & Band Lead

- Served as President and Band Lead of the college music club, overseeing club operations, organizing performances, and leading the band across college events and cultural fests.